

2 Antibiotics are medicines used to treat some medical conditions.

- (a) The table shows some medical conditions and whether or not antibiotics are needed to treat the condition.

Medical condition	Are antibiotics needed to treat the condition?
Impetigo	yes
Whooping cough	yes
Middle ear infections	sometimes
Sinus infections	sometimes
Multiple sclerosis	no
Rheumatoid arthritis	no

Explain the use of antibiotics to treat these conditions.

(2)

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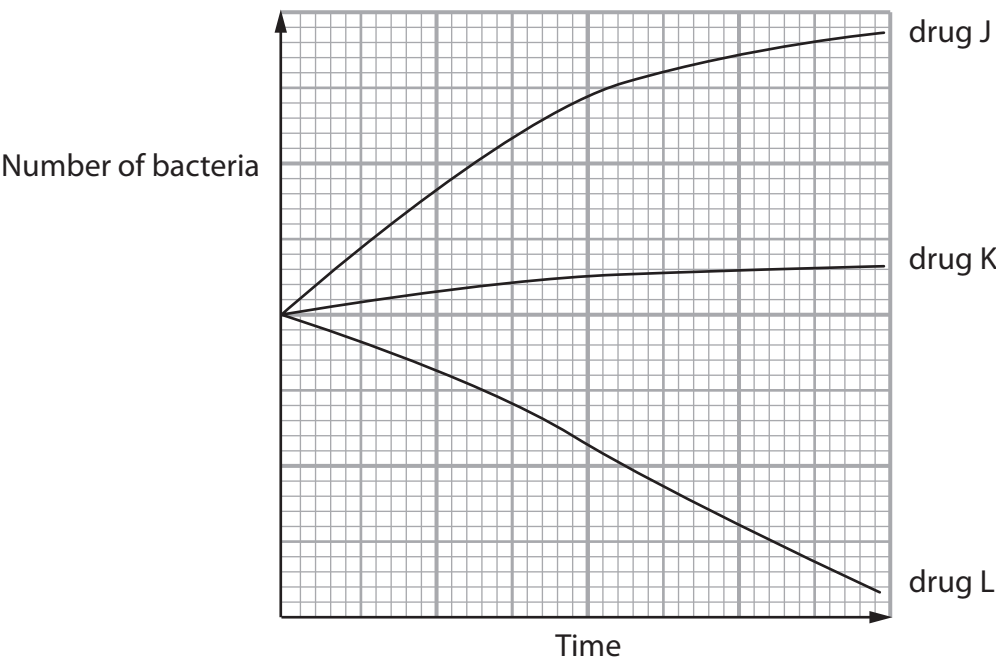
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(b) The graph shows the effect of three drugs, J, K and L, on the number of bacteria in a culture.



Which row of the table describes each of these three drugs?

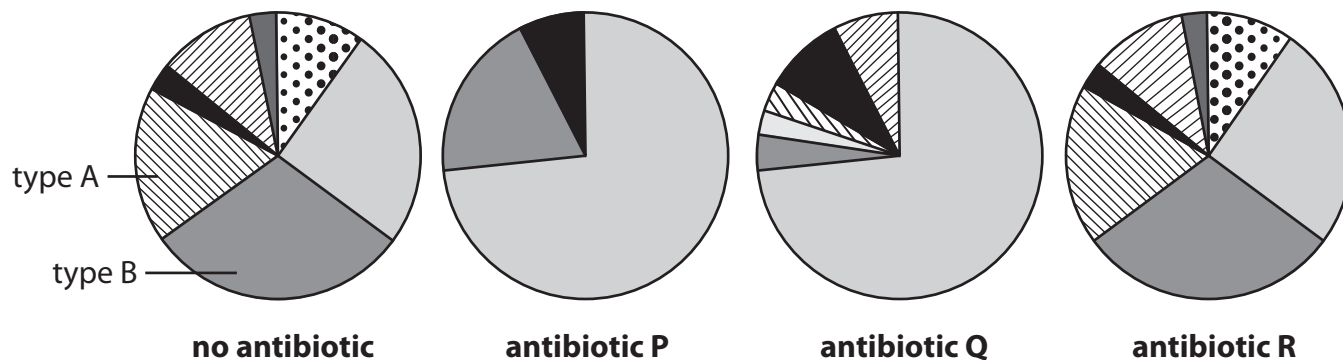
(1)

	drug J	drug K	drug L
☐ A	not an antibiotic	bactericidal antibiotic	bacteriostatic antibiotic
☐ B	not an antibiotic	bacteriostatic antibiotic	bactericidal antibiotic
☐ C	bactericidal antibiotic	not an antibiotic	bacteriostatic antibiotic
☐ D	bacteriostatic antibiotic	not an antibiotic	bactericidal antibiotic

(c) One problem of taking antibiotics is their effect on gut flora.

The diagrams show the effects of three antibiotics, P, Q and R, on the proportion of different types of gut flora.

Each section in each of the diagrams represents a different type of gut flora.



(i) Explain the role of gut flora in protecting the body from infection.

(2)

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(ii) Explain why antibiotics can affect gut flora.

(2)

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(iii) The number of type A in gut flora in the absence of antibiotics is 6 000 000.

Estimate the number of type B in gut flora in the absence of antibiotics.

Give your answer in standard form.

(1)

Answer

(iv) Deduce the effects of antibiotics P, Q and R on gut flora.

(3)

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(Total for Question 2 = 11 marks)